In particular,

$8! = 8\times(7\times6\times5\times4\times3\times2\times1)$

$= 8\times7!$

Example 2.8

Evaluate the following:

(i) $\frac{7!}{6!}$ (ii) $\frac{8!}{5!}$ (iii) $\frac{9!}{6!3!}$

Solution

(i) $\frac{7!}{6!} = \frac{7\times6!}{6!} = 7$

(ii) $\frac{8!}{5!} = \frac{8\times7\times6\times5!}{5!} = 336$

(iii) $\frac{9!}{6!3!} = \frac{9\times8\times7\times6!}{6!\times3!} = \frac{9\times8\times7}{3\times2\times1} = 84$

Example 2.9

Rewrite 7! in terms of 5!

Solution

$7! = 7\times6!$

$= 7\times6\times5! = 42\times5!$

Example 2.10

Find n, if $\frac{1}{9!}+\frac{1}{10!}=\frac{n}{11!}$

Solution

$$\frac{1}{9!}+\frac{1}{10!} = \frac{n}{11!}$$

$$\frac{1}{9!}+\frac{1}{10\times9!} = \frac{n}{11!}$$

$$\frac{1}{9!}\left[1+\frac{1}{10}\right] = \frac{n}{11!}$$

$$\frac{1}{9!}\times\frac{11}{10} = \frac{n}{11!}$$

$$n = \frac{11!\times11}{9!\times10}$$

$$= \frac{11!\times11}{10!}$$

$$= \frac{11\times10!\times11}{10!}$$

$$n = 121$$

2.2.2 Fundamental principle of counting

Multiplication principle of counting:

Consider the following situation in an auditorium which has three entrance doors and two exit doors. Our objective is to find the number of ways a person can enter the hall and then come out.

Assume that, P_1, P_2 and P_3 are the three entrance doors and S_1 and S_2 are the two exit doors. A person can enter the hall through any one of the doors P_1, P_2 or P_3 in 3 ways. After entering the hall, the person can come out through any of the two exit doors S_1 or S_2 in 2 ways.

Hence the total number of ways of entering the hall and coming out is

$3 \times 2 = 6$ ways.

These possibilities are explained in the given flow chart (Fig. 2.1).

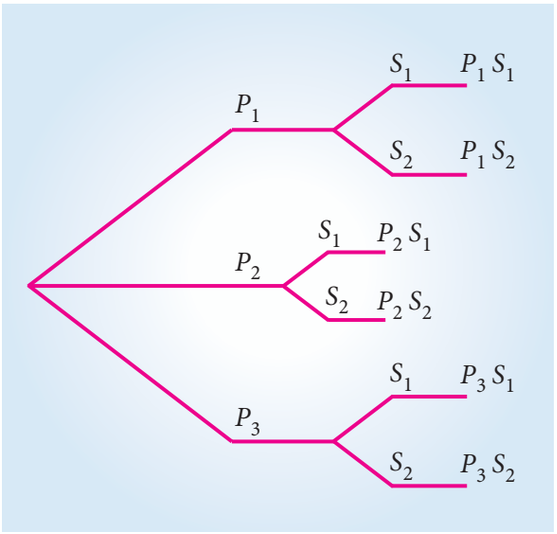


Fig. 2.1

The above problem can be solved by applying multiplication principle of counting.

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